

Volga Can Karakus

Full-Stack & Embedded Software Engineer, Agricultural & Biological Engineer

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SUMMARY

Software engineer with an off-road equipment engineering background and 9+ years building end-to-end systems for precision agriculture: field robotics, sensing, and the data pipelines behind them. From microcontroller firmware to cloud analytics pipelines and the mobile and web apps on top of them. Comfortable across the full stack: embedded C/C++, Python data and computer-vision pipelines, containerized microservices on Kubernetes, and native/cross-platform mobile development. I take products from sensor to dashboard. Currently also shipping a solo-built Unity mobile game, and I work day-to-day with agentic AI tooling (Claude Code) as a core part of my development loop.

TECHNICAL SKILLS

Languages	Python, C, C++, C#, JavaScript / TypeScript, Bash / Shell, Java, Swift, Dart, HTML/CSS, SQL
Cloud & DevOps	AWS (S3, EC2, Kinesis/KVS), Azure (AKS, Blob Storage), GCP, Docker & Compose, Kubernetes, Firebase
Backend & Data	Node.js / Express, Flask, Meteor, MongoDB (incl. Realm & change streams), Redis, REST, WebSockets, JWT
Mobile	SwiftUI (iOS), React Native / Expo, Flutter, Android (Java)
Frontend	React, Svelte / SvelteKit, Vite, D3.js
Robotics / ML	ROS2 (Humble), LiDAR processing, OpenCV, YOLOv8 / Ultralytics, NumPy / SciPy, GIS (GeoPandas, Shapely, PyProj, Rasterio, QGIS)
Embedded	PlatformIO, ESP32 / ESP32-CAM, Adafruit Feather M0 (SAMD21), FreeRTOS, BLE, LoRa
Game Development	Unity 6 (URP) & C#, procedural mesh generation, HLSL / ShaderLab, code-built UI, save & offline-progression systems, Unity Test Framework, iOS & Android build pipelines
AI-Assisted Dev	Claude Code (agentic development loops, custom skills / slash commands, subagents, hooks)

EXPERIENCE

Senior Software Engineer | [EarthSense](#)

2018 – 2026

Agricultural Robotics — TerraSentia Platform

Full-stack contributor across the TerraSentia autonomous field-robot platform, owning systems from firmware up through the cloud analytics that turn raw collections into crop data.

Built and operated a containerized **cloud analytics pipeline** on Azure Kubernetes (AKS) that turns raw LiDAR and camera data into per-plot crop phenotypes (**height, leaf area index, width, stand count, point clouds**), delivered as a table mirroring each grower's field layout; jobs orchestrated via MongoDB change streams and a Redis queue across AWS, Azure and GCP storage.

Implemented the **LiDAR crop-height algorithm** Turn the sensor readings into map coordinates, trace the left and right boundaries, take a robust average while ignoring outliers. Authored reusable internal Python libraries for odometry, camera projection, point clouds and cloud storage.

Developed an **autonomous path generator** producing serpentine field routes that complies with the restriction of the vehicle movements, using computer vision to detect tree centres and field boundaries — UTM projection, row clustering and curvature-constrained Bézier U-turns, exported as GeoJSON for GIS tools.

Created a **web-based teleoperation system** for remote robot control: real-time video streaming and autonomous waypoint navigation over WebSockets and AWS Kinesis/KVS, integrated with ROS2 Humble and images are pushed to Azure Container Registry for over-the-air updates to robots in the field.

Built the internal **data-review web app** (SvelteKit + Vite, Firebase hosting, MongoDB Realm backend, Azure identity / JWT) for plot alignment, video/LiDAR review and launching analytics pipelines.

Wrote **LoRa communication firmware** (C++/PlatformIO) for Adafruit Feather M0 and custom modules, and maintained the **Android operator apps** used to drive the robots in-field.

Founder & Solo Developer | [Idle ShroomVerse](#)

2025 – Present

Personal Product — Unity Mobile Game (iOS & Android)

Designed and built, as a solo developer, a mobile strategy game where players grow colonies across procedurally generated planets, earn progress while away, and raid rival players. Built end to end with Claude Code, from design through testing and release.

Technologies: Unity, C#, HLSL, Python, Claude Code, MeshyAI MCP

Generated worlds: every planet is created automatically from a single number, so the same planet looks identical every time. Wrote custom visual effects and made planets that can be shattered into pieces, leaving debris rings in orbit.

Hand-built interface: coded all ~30 menus and screens from scratch, adapting to different phone screen shapes.

Progress while away: when players return, the game replays what would have happened using the same rules as live play, so offline results match real play. Saves are protected against corruption and stay compatible with future updates.

Player-vs-player raids: players can scout and attack each other even when the other player is offline, with protection against being attacked too often. Backed by 69 automated tests.

Game balance: modeled the economy and battles in Python before building them, then tuned the game's progression to match.

Performance: a colony can simulate thousands of workers while only drawing a few hundred on screen, keeping the game smooth on phones.

AI-assisted development: used Claude Code as the main coding tool, guided by a detailed design document that kept changes consistent across a 200-file codebase.

Software Engineer | [Pro Bono](#)

2026

Temporary help provided

Developed a computer-vision pest-detection service — a YOLOv8/OpenCV model (Dockerized) for automated detection and counting of Mediterranean fruit fly (*Ceratitis capitata*) from trap imagery.

Personal Project & Solo Engineer | [Eden \(EdenBox\)](#)

2024 – Present

Personal Product — Smart Garden Sensor

Designed an end-to-end consumer IoT product: **ESP32 firmware** (PlatformIO/C++) handling BLE setup, low-frequency ESP32-CAM image capture and capacitive/SHT moisture sensing.

Built the companion **SwiftUI iOS app** (MVVM + Combine) with BLE device provisioning, Supabase authentication, sensor dashboards and an image gallery — plus React Native/Expo and Flutter client prototypes over a Supabase backend.

SELECTED PROJECTS

Robot Data Webapp | Node.js, Express, React, MongoDB

JWT-authenticated dashboard visualizing robot field metrics (node and ear counts) with interactive charts.

Companion Logging Web | Node.js, Docker, MongoDB

Local Dockerized site with credential encryption and verification against MongoDB, surfacing on-device robot collections.

John Deere — Tango Lawnmower Automation | Android, Embedded Wi-Fi

Senior capstone: Android controller for an automated mower over a self-programmed Wi-Fi module, with CAD/3D-printed hardware; presented to John Deere engineers.

EDUCATION

University of Illinois at Urbana-Champaign

May 2018

B.S. in Agricultural Off-Road Equipment Engineering, Minor in Computer Science

Coursework: Robotics & Manipulation, Field Robotics (Sensing), Embedded Systems (Teaching Assistant), Artificial Intelligence, Computer Graphics, Mechatronics, GIS & Data Analysis, Engineering Measurement Systems, Dynamic Mechanics (Teaching Assistant).

Lycée Pierre Loti d'Istanbul

May 2013

French Baccalaureate, Science Specialization — Istanbul, Turkey